

FireFair: Equity-Adjusted Multi-Agent Triage for Wildfire Ignition Forecasting

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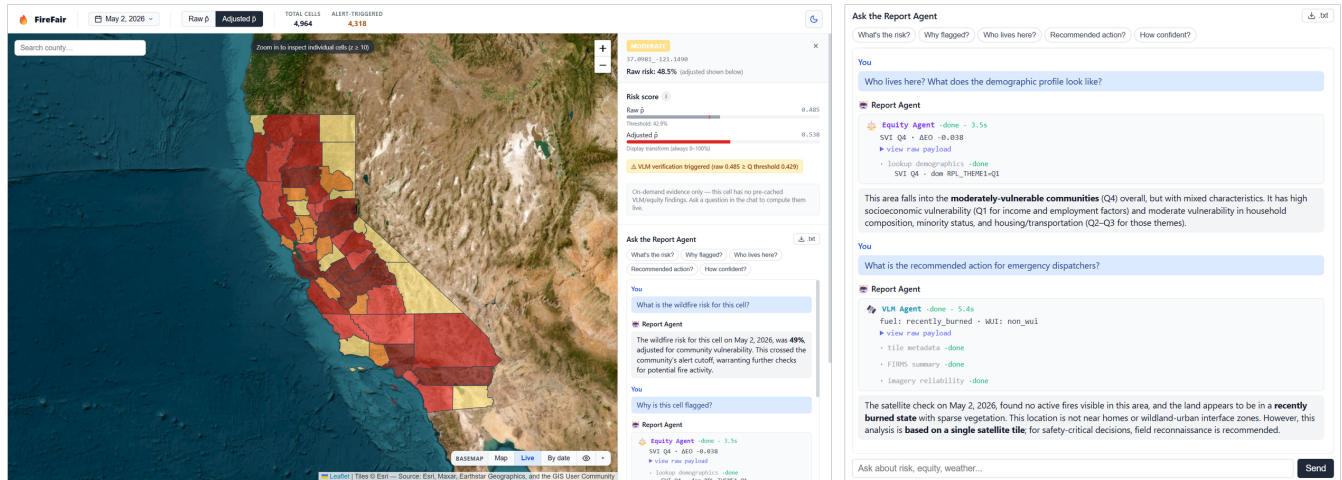


Figure 1: FireFair prototype: equity-adjusted California risk map (left) and multi-agent chat panel with visible tool-call traces (right). Map image © 2026 Esri and its licensors, used under license.

Abstract

Machine learning-based wildfire ignition forecasting is increasingly applied to inform emergency response, yet aggregate metrics such as F_1 and AUROC can mask systematic under-warning of socially vulnerable communities. We present FIREFAIR, a three-layer system that audits a wildfire forecaster by the CDC/ATSDR Social Vulnerability Index (SVI), a composite measure of socioeconomic and demographic risk factors at the census-tract level, then combines

(i) a score-level equity-aware recall adjustment inspired by equal-opportunity post-processing, (ii) per-community routing gates, and (iii) a chat-based multi-agent verification layer whose tool calls are visible to dispatchers. On a held-out California test fold, the audit reveals systematic under-warning, with an equal-opportunity gap of -0.155 between the most- and least-vulnerable quintiles on the evaluation half ($n=3,579$). The score-level adjustment, expressed as a single auditable equity-adjusted score, reduces this disparity by 75% (to -0.038 ; 95% bootstrap CI crosses zero) at a 1.6 pp aggregate F_1 cost. On the 2025 Madre Fire, this cutoff promotes 36% of marginal-signal cell-days into multi-agent verification (including a single-cell 2-day pre-ignition lead), where the agent stack corroborates the signal and surfaces tract-level racial/ethnic minority status through visible tool calls. A prototype is available at <https://fire-fair.github.io/>.

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GoodIT '26, Pisa, Italy
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ACM ISBN 979-8-4007-2483-1/26/09
<https://doi.org/10.1145/3794786.3830751>

CCS Concepts

• **Applied computing** → **Earth and atmospheric sciences**; • **Social and professional topics** → **Computing / technology policy**; • **Computing methodologies** → **Multi-agent systems**; *Supervised learning*.

Keywords

algorithmic fairness, equity-stratified evaluation, wildfire ignition forecasting, social vulnerability index, multi-agent LLM systems, environmental justice, computing for social good

ACM Reference Format:

Kefei Yan, Frank F. Yang, Yiyang Wang, Ziyi Yang, Kewen Liu, Yuzhuo Chen, Alex Cabral, and Josiah Hester. 2026. FireFair: Equity-Adjusted Multi-Agent Triage for Wildfire Ignition Forecasting. In *International Conference on Information Technology for Social Good (GoodIT '26), September 02–04, 2026, Pisa, Italy*. ACM, New York, NY, USA, 6 pages. <https://doi.org/10.1145/3794786.3830751>

1 Introduction

California wildfires drive escalating harms: smoke-driven respiratory and cardiovascular illness [5], ecosystem loss [17], and devastation concentrated in communities least able to recover. For example, the 2018 Camp Fire destroyed the entire town of Paradise and killed 85 people [21]. The January 2025 Palisades and Eaton fires jointly caused an estimated \$95–\$164 billion in losses [19], with the highest-burden communities experiencing the greatest smoke exposure [16]. As ML-based wildfire ignition forecasting is increasingly applied to inform emergency response and resource pre-positioning [23, 30], an open question remains: to what extent does the aggregate accuracy of these models mask systematic under-warning of those most at risk?

The convention in wildfire ML evaluation does not account for this potential inequity: recent benchmarks report aggregate F_1 and AUROC averaged across geography [9, 23, 30]. Such summary metrics, by construction, cannot reveal whether a forecaster systematically under-warns tracts flagged by the Centers for Disease Control and Prevention and Agency for Toxic Substances and Disease Registry (CDC/ATSDR) Social Vulnerability Index (SVI) [4]. The absence of equity-focused evaluation in wildfire ML mirrors a broader pattern in high-stakes ML: predictive policing systems amplify racially disparate enforcement through feedback loops on biased training data [20], and medical risk scores have systematically under-referred Black patients for care [25]. To our knowledge, no analogous audit has been performed for wildfire ignition prediction. If models systematically miss fires in disadvantaged communities, the technology may amplify existing inequity rather than reduce it.

Furthermore, wildfire triage combines heterogeneous data sources such as weather-driven forecast scores, vulnerability tract data, and satellite imagery, each governed by distinct trust, update frequency, and cost properties. A single end-to-end model conflates these signals in ways that place the equity audit after the prediction step rather than within the prediction process itself.

To address these issues, we present **FireFair**, a three-layer system that combines a single ignition probability with an equity audit and multimodal verification, presented to the dispatcher as a chat-based brief. Built on an InceptionTime+CutMix California ignition

forecaster, FireFair routes high-risk grid cells through a Vision-Language Model (VLM) Agent for Sentinel-2 evidence, an Equity Agent for SVI-stratified auditing, and a Report Agent whose brief is explicitly conditioned on equity findings. Our contributions are:

- **Audit.** We perform an equity-stratified evaluation of an off-the-shelf wildfire ignition forecaster on CDC/ATSDR SVI quintiles, instantiating two fairness statistics for spatial hazard prediction: the equal-opportunity gap Δ_{EO} [13] and the alert selection-rate ratio ρ [12].
- **Method.** We operationalize Hardt et al.’s equal-opportunity post-processing as a single score-level adjustment (Eq. (1)) that composes with downstream agentic routing rather than requiring per-group threshold tables.
- **System.** We build and deploy this system as a prototype validated forward-in-time on the 2025 Madre Fire. The equity-adjusted cutoff routes 36% of marginal-signal cell-days into multimodal VLM verification (including a single-cell 2-day pre-ignition lead), and the Equity Agent surfaces tract-level vulnerability through visible tool-call traces, making fairness a first-class input rather than a post-hoc filter.

2 Related Work

Wildfire ML benchmarks. Researchers have recently begun to advance the wildfire forecast modeling frontier with new datasets and sequence architectures [9, 23, 30]. However, these models evaluate only at aggregate geographic scale, leaving open *whose* communities these models serve once they are used for operational triage.

Agentic and multimodal hazard systems. Multi-agent and VLM-based hazard pipelines use retrieval, chain-of-thought oracles, and geospatial grounding to combine heterogeneous data [6, 18, 22, 29], but they treat demographic context as descriptive metadata. FireFair instead makes the audited fairness signal a routing input and reserves the agent layer for visual interpretation, adaptive equity context, and dispatcher-facing explanation.

Environmental justice and spatial fairness. Empirical work documents disparate wildfire exposure across socially vulnerable populations [24], allocative tradeoffs in environmental-justice tools [14], and design frameworks treating equity as a first-class requirement rather than a post-hoc consideration [7, 8, 26]. FireFair operationalizes these concerns within a live hazard-response pipeline.

3 System Overview

Figure 2 shows the end-to-end architecture. FireFair is organized as three composable layers: **Layer 1** (score-level equity adjustment, §4.2) and **Layer 2** (per-community routing gates, §4.3) run inside the always-on backbone (left column of Figure 2); **Layer 3** (chat-based multi-agent verification, §3.2) is invoked on demand by the Report Agent and its sub-agents (right side of Figure 2). The daily scheduler runs only deterministic components: it ingests gridMET weather features, scores the InceptionTime ensemble to obtain a raw ignition probability $\hat{p}$, applies the Layer-1 recall adjustment to obtain the equity-adjusted score $\hat{p}$, and stores the resulting cell-level predictions. LLM calls (Layer 3) are triggered on demand by user interaction or by server-side Layer-2 routing gates, keeping the

expensive, stochastic agent layer off the critical batch path while preserving an auditable interface for high-risk cells.

3.1 Dataset

Backbone data and splits. We train the InceptionTime ensemble on the California subset of the FireCastRL US Wildfire Dataset [23], where each *cell* is a ~4 km gridMET grid square and a *cell-day* is one cell on one date. The dataset provides 9.5 M CONUS cell-days (January 2014 to April 2025) as 75-day sequences over 15 daily gridMET variables (weather, fuel-moisture, and water-balance categories), paired with next-day ignition labels. Preserving FireCastRL’s stratified 80/10/10 split and restricting to California yields $n_{\text{test}} = 7,154$ test cell-days. To avoid leakage when estimating the per-stratum recall $\hat{r}_s$ (§4.2), we further split this test fold into an SVI-stratified *calibration half* ($n_{\text{cal}} = 3,575$, used only for $\hat{r}_s$) and *evaluation half* ($n_{\text{eval}} = 3,579$, used for all reported metrics).

Equity stratification. Each cell is joined by centroid to its 2020 census tract and assigned the CDC/ATSDR Social Vulnerability Index (SVI) 2022 quintile [4] ($s=1$: most vulnerable), which aggregates 16 American Community Survey variables across four themes (socioeconomic status, household composition, racial/ethnic minority status, and housing/transportation).

Multimodal and case-study data. For visual verification, the VLM Agent consumes ESA Sentinel-2 Level-2A RGB tiles (10 m, ~5-day revisit) [10] composited with NASA/NOAA Visible Infrared Imaging Radiometer Suite (VIIRS) 375 m active-fire detections [27] delivered through NASA FIRMS;¹ tiles are pulled only after the Layer-2 alert gate fires (§4.3). The two forward-time case studies, the 2025 Madre Fire (§5.2) and Shasta hotspot (§5.3), both fall after the backbone’s April 2025 training cutoff and the LLM’s October 2023 knowledge cutoff; ground truth is sourced from CAL FIRE incident pages [1–3] and InciWeb [28].

3.2 Agents and Tools

A monolithic LLM would mix evidence streams of very different provenance, latency, and audit cost; FireFair’s Layer 3 assigns each stream to a specialized agent with its own tool budget and gating rule, so every step of the evidence chain is inspectable in the chat trace. The Report Agent is the user-facing orchestrator, with five tools covering risk lookup, weather trends, equity audit, visual analysis, and nearby FIRMS hotspots. The Equity Agent joins a cell to its SVI tract and applies sample-size, effect-size, and interval gates before flagging a disparity. The VLM Agent reads a Sentinel-2 RGB tile and a VIIRS thermal-anomaly summary, returning typed evidence for fuel state, active fire, wildland–urban interface proximity, and uncertainty. Tool calls stream as Server-Sent Events, so the dispatcher sees the evidence path, not just the answer. All three agents use OpenAI function-calling (native tool use) on gpt-4o (snapshot 2024-11-20; knowledge cutoff October 2023).

¹<https://firms.modaps.eosdis.nasa.gov/>

4 Method

4.1 Backbone Predictor

The predictor is a five-member InceptionTime ensemble [15] trained with CutMix regularization [31] on the California gridMET sequences (§3.1). The raw ignition probability $\hat{p}_i$ is the average positive-class softmax across the five members, reaching $F_1 = 0.741$ on a stratified evaluation half ($n_{\text{eval}} = 3,579$), comparable to the 0.7–0.8 range reported on related North American wildfire panels in recent work [23, 30]. The architecture is intentionally off-the-shelf; FireFair’s contribution is the equity-aware scoring (§4.2), routing (§4.3), and agent stack (§3.2) that wrap this backbone.

4.2 Layer 1: Equity-Adjusted Scoring

To reduce the per-stratum recall gap of the raw backbone, Layer 1 rescales $\hat{p}_i$ into an equity-adjusted score $\tilde{p}$ that lowers routing cutoffs for historically under-recalled strata. Each cell carries an SVI quintile $s(i) \in \{1, \dots, 5\}$ ($s=1$: most vulnerable), and the test fold is split into calibration ($n_{\text{cal}} = 3,575$) and evaluation ($n_{\text{eval}} = 3,579$) halves (§3.1). At the baseline threshold $\tau = 0.5$, the empirical per-quintile recall on the calibration half is $\hat{r}_s = (0.687, 0.700, 0.697, 0.858, 0.908)$ for Q_1 – Q_5 , a 22-point gap between the most and least vulnerable quintiles. Because $\hat{r}_s$ is estimated on the calibration half and all reported mitigation metrics are computed on the disjoint evaluation half, the adjustment does not use evaluation labels.

FireFair then computes the equity-adjusted routing score

$$\tilde{p}_i = \frac{\hat{p}_i}{\max(\hat{r}_{s(i)}, r_{\min})}, \quad r_{\min} = 0.5. \quad (1)$$

For a fixed threshold τ , Eq. (1) is algebraically equivalent to the raw-score cutoff $\hat{p}_i \geq \tau \max(\hat{r}_{s(i)}, r_{\min})$: groups with lower historical true-positive rates receive lower cutoffs for follow-up triage, a deterministic score-level operationalization inspired by equal-opportunity post-processing [13]. The contribution is architectural rather than theoretical: instead of Hardt et al.’s per-group threshold tables, Eq. (1) packages the same post-processing as a single auditable score $\tilde{p}$ that composes with downstream routing gates, trading some flexibility in the post-processing geometry for compositional audibility across Layers 2 and 3. In the deployed UI, $\tilde{p}$ is displayed via a bounded monotone transform into $[0, 1]$; routing decisions still use $\tilde{p}_i$ itself.

4.3 Layer 2: Routing and Fairness Metrics

All routing gates operate on $\tilde{p}$: a cell enters the high-risk set when $\tilde{p}_i \geq \tau$ and is sent to the VLM when $\tilde{p}_i \geq \tau_v$. The current deployment uses $\tau = \tau_v = 0.5$ and enforces the VLM gate as a server-side hard rule, operationalized as the algebraically equivalent per-community cutoff on the raw score, $\hat{p}_i \geq \tau_v \cdot \max(\hat{r}_{s(i)}, r_{\min})$, so the decision references a bounded quantity even when $\tilde{p}_i$ exceeds one. At this setting, the resulting per-quintile cutoffs on $\hat{p}_i$ are (0.343, 0.350, 0.349, 0.429, 0.454) for Q_1 – Q_5 .

We report group-conditional recall within each SVI quintile and two summary metrics: the equal-opportunity gap [13]

$$\Delta_{\text{EO}} = \text{Recall}_{Q_1} - \text{Recall}_{Q_5}, \quad (2)$$

and the selection-rate ratio $\rho = (a_1/n_1)/(a_5/n_5)$ from disparate-impact analysis [12], where a_k and n_k count alerts and cells in

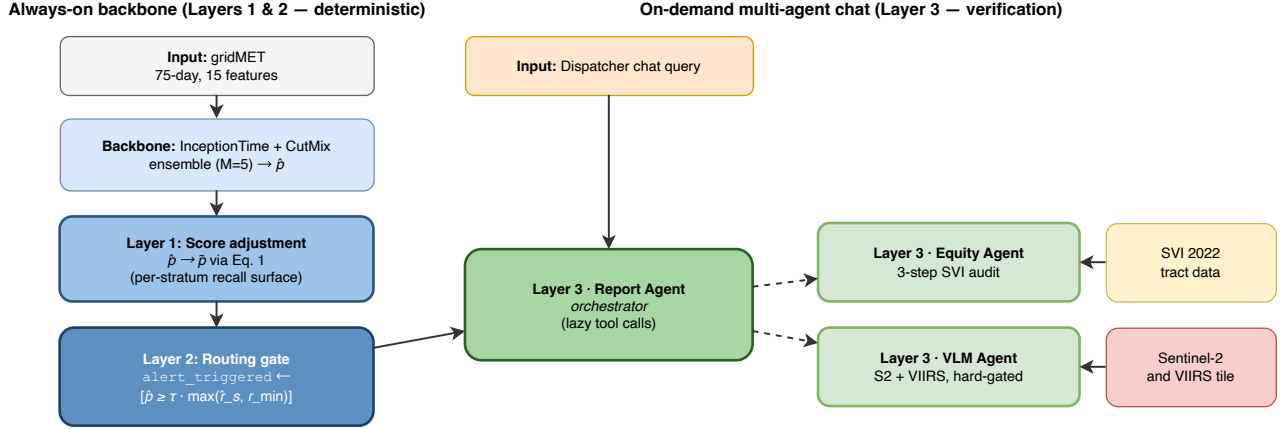


Figure 2: FireFair architecture, organized as three layers. Solid arrows: deterministic data flow. Dashed arrows: the Report Agent’s selective tool calls (Equity Agent on demand; VLM Agent hard-gated on alert_triggered). Left (Layers 1 & 2, always-on backbone): gridMET sequences feed the InceptionTime+CutMix ensemble to produce $\hat{p}$; Layer 1 rescales $\hat{p}$ into $\tilde{p}$ via Eq. (1); Layer 2 emits alert_triggered. Right (Layer 3, on-demand verification): the Report Agent receives the cell context and lazily invokes the Equity Agent over SVI 2022 tract data and, when gated, the VLM Agent over a Sentinel-2 and VIIRS tile.

quintile k . Negative Δ_{EO} and $\rho < 1$ both indicate under-warning of the most vulnerable quintile. Confidence intervals use 2,000 percentile-bootstrap [11] resamples of the evaluation half.

5 Evaluation

We present preliminary evidence for FireFair along two axes: a held-out equity-adjusted triage ablation (§5.1) and two forward-time 2025 California fire case studies illustrating complementary operational modes (§5.2–§5.3).

5.1 Equity-Adjusted Triage Ablation

Table 1 contrasts raw thresholding with FireFair routing on the held-out evaluation half ($n_{\text{eval}} = 3,579$). The adjusted configuration reduces $|\Delta_{EO}|$ from 0.155 to 0.038 (75% reduction; adjusted Δ_{EO} ’s 95% bootstrap CI $[-0.084, +0.006]$ crosses zero). The selection-rate ratio ρ rises from 0.71 to 0.88, narrowing the alert-rate gap as a routing-side byproduct of the recall adjustment rather than as a separate fairness target. Operationally, the adjusted routing’s recall gap is statistically indistinguishable from zero on this fold: the disparity that the audit surfaced in the raw backbone no longer survives the routing stage.

Aggregate F_1 falls by 1.6 pp; review load rises by 22% (1,582 → 1,933 cells), with Q_1 review counts up 26% (Table 1). The per-quintile recall lift concentrates in the most vulnerable quintiles (+13.1, +15.8, +10.1, +3.8, +1.4 pp for Q_1 – Q_5): the additional review budget routed through the agent stack lands on exactly the strata the raw backbone was under-recalling, not on Q_4/Q_5 cells that already alert reliably. The $r_{\min} = 0.5$ floor is inactive on this data, since every observed $\hat{r}_s \in [0.69, 0.91]$. The 75% disparity reduction is a deterministic consequence of the per-stratum cutoff geometry (§4.2), not an optimization result; a naive uniform lowest-threshold ($\tau = 0.343$ everywhere) would over-trigger on Q_4/Q_5 where $\hat{r}_s$ is already high, inflating review load without improving Q_1 recall.

Table 1: Held-out triage ablation ($n_{\text{eval}} = 3,579$). Δ_{EO} reports $\text{Recall}_{Q_1} - \text{Recall}_{Q_5}$ with 95% bootstrap CI over 2,000 resamples; review counts use the per-community cutoffs of §4.3, equivalently $\tilde{p}_i \geq 0.5$.

Config.	F_1	Δ_{EO}	95% CI	ρ	Q_1	Total
Raw $\hat{p}$	0.741	−0.155	[−0.216, −0.098]	0.71	301	1,582
Adj. $\hat{p}$	0.725	−0.038	[−0.084, +0.006]	0.88	380	1,933

In practice, FireFair adds approximately 350 cell-day reviews per evaluation window to a dispatcher’s queue, concentrated in Q_1 cells the raw threshold would have silently filtered out.

5.2 Forward-Time Case Study: 2025 Madre Fire

The 2025 Madre Fire ignited in northern San Luis Obispo County on 2025–07–02 and burned 80,779 acres before being declared out on 2025–07–26 [2, 28]. Because the fire post-dates the backbone’s April 2025 training horizon, it provides a forward-time illustration of the system’s auditability pathway: we trace the deterministic routing decision (Layer 1+2) and the on-demand agent verification (Layer 3) cell-day by cell-day. We take a 21-day window spanning 2025–06–25 to 2025–07–15 that brackets the official ignition date, and select the five gridMET cells whose centroids fall within 20 km of the CAL FIRE perimeter (105 cell-days total). The fire centroid sits in SVI Q_1 [4], with the Equity Agent identifying racial/ethnic minority status as the dominant vulnerability theme (Figure 3).

Layer 1+2: marginal-signal rescue. Across 105 cell-days, the raw threshold $\hat{p} \geq 0.5$ issues *zero* alerts. The equity-adjusted cutoff (Q_1 value 0.343) lets 38/105 (36.2%) cell-days cross into the multi-agent verification path; the threshold relaxation alone does not change $\hat{p}$, and the rescue depends on multimodal confirmation downstream. Under raw thresholding the Madre Fire would have produced no

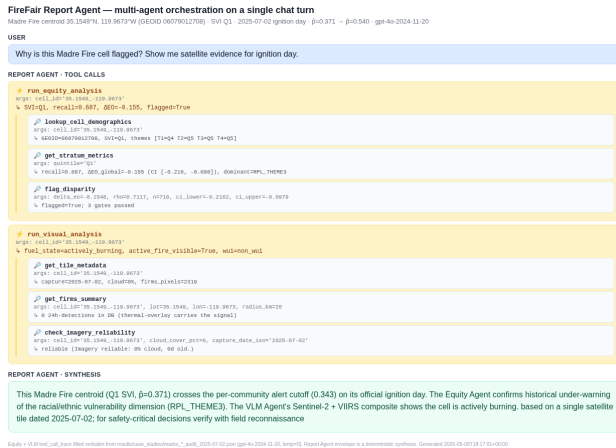


Figure 3: FireFair multi-agent collaboration on the Madre Fire centroid (SVI Q_1 ; ignition day 2025-07-02; $\hat{p} = 0.371 \rightarrow \tilde{p} = 0.540$). The Report Agent invokes the Equity Agent (3-step adaptive audit) and the VLM Agent (3 sub-tools, mandatory uncertainty caveat), then synthesizes a brief conditioned on the equity flag. Tool calls stream live as Server-Sent Events.

automatic VLM review across the entire 21-day window; the equity adjustment converts a silent miss into a multi-agent triage event that a dispatcher can inspect. At the centroid on ignition day, $\hat{p} = 0.371$ clears the Q_1 cutoff while missing the raw threshold (equivalently $\hat{p} = 0.540 \geq 0.5$). *Two days before ignition*, $\hat{p} = 0.359$ already clears the cutoff ($\hat{p} = 0.523$), opening a plausible pre-ignition review window on the routing score alone.

Layer 3: multi-agent collaboration on the ignition day. Figure 3 shows the production chat trace from our prototype on the Madre centroid for ignition day 2025-07-02. Answering the dispatcher query “Why is this Madre Fire cell flagged?”, the Report Agent lazily invokes two sub-agents. The Equity Agent (3.5 s, three tool calls) runs the full adaptive audit, identifying SVI Q_1 with a dominant racial/ethnic minority theme and a per-stratum recall of 0.687 ($\Delta_{EO} = -0.155$, 95% CI $[-0.216, -0.098]$); on a non-extremal adjacent cell the same prompt short-circuits at the demographic lookup, demonstrating a stratum-adaptive tool budget. The VLM Agent (5.4 s, three sub-tool calls) reads a Sentinel-2 and VIIRS composite (2,319 thermal pixels) and emits an actively-burning fuel state with a mandatory uncertainty caveat. The Report Agent synthesizes both into a dispatcher brief *conditioned on the Equity flag*, surfacing the tract’s racial/ethnic minority status rather than treating it as a footnote; tool calls stream as Server-Sent Events so the dispatcher sees the evidence path.

5.3 Shasta County: Adaptive Equity Audit

A second forward-time case provides a contrastive regime where the backbone alerts cleanly, so Layer 1+2 has no marginal signal to rescue and Layer 3 shifts from *verifying a rescue* to *adaptive within-fire audit*. The cell at (40.3863, -122.3427) in Shasta County coincides with a persistent VIIRS 375 m active-fire signature [27] peaking on

2025-06-19 and co-occurs with two officially recorded CAL FIRE Shasta-Trinity Unit incidents the following day: the Happy Fire in Happy Valley and the Rush Fire south of Lewiston in adjacent Trinity County [1, 3]. The cell sits in SVI Q_1 and shows $\hat{p} = 0.627$ and $\hat{p} = 0.913$ on the peak day, with both raw and adjusted scores firing across the 10-day window; equity adjustment here normalizes priority rather than rescuing missed alerts.

The agent contribution surfaces in a within-window contrast. On the Q_1 centroid, the Equity Agent runs its full three-step audit and flags a racial/ethnic-minority disparity in recall; on an adjacent Q_4 cell ~6 km NE the same prompt short-circuits at the demographic lookup, since the agent’s decision tree routes non-extremal strata directly to synthesis. The tool budget thus adapts to community vulnerability rather than being expended uniformly: Layer 3 turns the equity signal into a routing decision inside the agent itself, not a downstream report a dispatcher has to read separately. The VLM Agent then reads the cloud-free Sentinel-2 and VIIRS composite (194 visible thermal pixels) and reports an actively-burning fuel state at the wildland–urban interface, multimodally confirming the signal. Madre and Shasta together illustrate two complementary modes: Layer 1+2 calibrated rescue for marginal signals, and Layer 3 adaptive tool budget when the backbone alerts cleanly.

6 Discussion, Limitations, and Future Work

From audit to intervention. The central design move in FireFair is to embed the equity adjustment in the routing layer itself, so that fairness operates as an operational signal rather than a reporting artifact. Our findings highlight complementary roles across the three FireFair layers in turning an audit into actionable intervention. The agent layer is more than a post-hoc explanation interface: each agent produces typed, inspectable evidence, and the dispatcher sees the full chain rather than a synthesized recommendation.

Limitations and next steps. As a work in progress, four caveats bound the current evidence and define our immediate research agenda. (i) The audit covers California only; cross-state transfer to additional U.S. wildfire panels is a natural next step. (ii) SVI is one vulnerability index; sensitivity analyses with alternative composite indices will test robustness to the choice of vulnerability metric. (iii) Broader prospective evaluation on recent CONUS fires remains the natural next milestone. Under the distribution shift well-documented in forward-time fire ML, the system’s value is *auditability* rather than statistical lift: the per-community cutoff makes the routing decision inspectable, and the agent stack surfaces evidence (or its absence) to the dispatcher rather than silently filtering. (iv) The VLM Agent inherits vision-language model hallucination risks; we mitigate with typed outputs and mandatory caveats, but operational deployment requires human-factors evaluation with emergency managers as future work.

7 Conclusion

FireFair operationalizes equity-aware wildfire triage as a three-layer system: score-level recall adjustment, per-community routing, and chat-based multi-agent explanation. On a stratified California evaluation half it reduces the SVI Q_1 – Q_5 equal-opportunity gap by 75% at a 1.6 pp F_1 cost. This cost buys not a more accurate backbone but wider multi-agent verification access for under-served

strata. On the 2025 Madre Fire the equity-adjusted cutoff promotes 36% of cell-days into the VLM verification gate (including a single-cell 2-day pre-ignition lead), where the agent stack corroborates the signal and surfaces tract-level racial/ethnic minority status through visible tool calls. A second Shasta County case shows the same agent stack shifting to adaptive within-fire audit when the backbone alerts cleanly. Ongoing work targets cross-state transfer, alternative vulnerability indices, and human-factors evaluation, toward operational deployment beyond California. The system advances UN SDG 10 (reduced inequalities), SDG 11 (sustainable cities and communities), and SDG 13 (climate action).

Acknowledgments

During the preparation of this work the authors used OpenAI ChatGPT and Anthropic Claude to assist with code development and language editing. All technical content, experiments, and claims were designed and verified by the authors, who take full responsibility for the work.

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